Local Properties of Binary Images in Two Dimensions

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Abstract—Aspects of topology and geometry are used in analyzing continuous and discrete binary images in two dimensions. Several numerical properties of these images are derived which are "locally countable." These include the metric properties area and perimeter, and the topological invariant, Euler number. "Differentials" are defined for these properties, and algorithms are given. The Euler differential enables precise examination of connectivity relations on the square and hexagonal lattices. Easily computable binary image characterizations are introduced, with reference to a serial binary image processor (BIP) now being built. A precise definition of "localness" is given, and some implications for image computation theory are examined.

Index Terms—Binary images, connectivity, Euler number, hexagonal lattice, local properties, neighborhood analysis, perceptron, serial processors, square lattice, theory of computation, topology.

I. Introduction

THIS paper describes certain properties of binary images in two dimensions, both in the continuous plane and on the hexagonal and square lattices.[1] These numerical properties are *locally countable*; that is, a given image property is a sum of numbers, where each number is derived from data contained in a small piece or "cell" of the image. (Cells are simply connected, completely cover the image, are smaller than significant image details, are all measured by the same process, and are independent of each other for measurement purposes.)

Of particular interest is the fact that the topological property "Euler number" of a region of space is locally countable (Fig. 1). Euler number in two-space is simply the number of distinct objects, including points, lines, and blobs, minus the number of holes inside objects. In one dimension, this definition reduces to the number of distinct segments along a line.

Euler number is useful in image processing both because the topological counts in a given image are of intrinsic interest in characterizing and recognizing the image, and because Euler number is a key concept in examining connectivity relations of individual bits on lattices. This is discussed at length below.

In addition to Euler number, certain metric properties are locally computable, such as perimeter length and area. Useful image characterizations can be obtained by formulas involving these measurements.

The concept of summing local measurements is, of course, closely related to the perceptron idea. Local measurements of an image can be carried out by first-level elements, each of which is limited in "diameter," or the distance spread of its inputs, and in "order," which is the number of its inputs [1, p. 12]. The elements' binary outputs are weighted and summed. (Perceptrons include a threshold or decision step. We omit this, talking only of numerical image measurement. Hence our use of "order" and "diameter" characterizes a measurement, and not a predicate as in perceptron literature.)

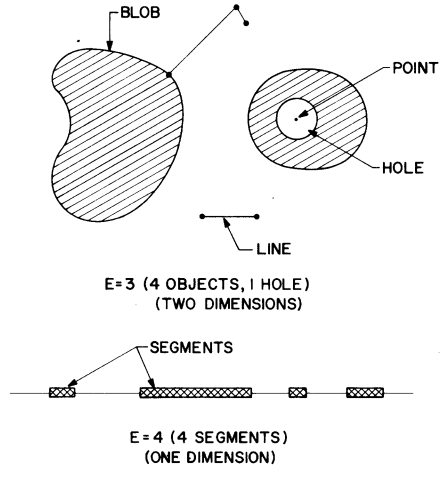


Fig. 1. Euler number in one and two dimensions.

This study of local measurements was prompted by a special computer, the binary image processor (BIP). The BIP examines small neighborhoods, one at a time, in two-dimensional binary images. For each neighborhood, several Boolean predicates are computed in parallel, based on various majority-type functions of bit cells and their immediate neighborhoods. Some of these predicates are counted, resulting in local property measurements; others can be used to derive new images, giving powerful image transformation capabilities.

II. Continuous Binary Images[2]

A. Definitions

A *border curve* on the plane is defined to be a simple bounded, closed curve. A *border set* is a finite set of border curves, none of which touch or intersect any others. *A binary region* is a finite, simply connected area on the plane, not crossed by any border curves, having assigned to it the image value 1 or 0. The *exterior* of the plane is that portion

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[1] "Two dimensions" is emphasized because all the concepts and methods in this paper extend readily to three-dimensional images. This extension will be covered in a subsequent paper.

[2] The material in this section is not new to topologists, but is presented here because it is difficult or impossible to find references which present all the required material in an elementary manner.

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away from and enclosing the region containing the border set. The exterior is a connected region because there is a finite number of bounded border curves. All of the exterior is assigned the image value 0.

The *parity* P of a point not on a curve, with respect to a given border set, is defined to be even or odd depending on whether a simple path from the point to the exterior, not tangent to any border curve, crosses an even or odd number of lines. It is not hard to show that P is independent of the path chosen.

Then a *binary image* is a plane containing a border set, and having an image value of 0 in every binary region whose points have even parity, and 1 everywhere whose points have odd parity. The parity and therefore the image value of two points separated by a single boundary curve are different, so a border curve always has image value 0 on one side and image value 1 on the other side. An *object border* is a border curve having image value 1 on the interior side of the curve. An *object* is the entire interior of an object border. A *hole border* is a border curve having image value 0 on the interior side of the curve. A *hole* is the entire interior of a hole border. (See Fig. 2.)

These definitions include only two-dimensional image objects and holes, because lower dimensional elements cannot occur on lattices, which is our main goal below; however they can be validly included in nonlattice discussions.

B. Image Triangulation

Suppose that in a given binary image we draw lines through the 1's areas, each ending at a border or at another line, such that all 1's regions thus created have three sides. See Fig. 3, which shows two different triangulations of a one-object image. Let n_0, n_1, and n_2, respectively, be the total number of vertices, line segments, and 1's regions in the image. Then the Euler number E of the image is defined as

$$E = n_0 - n_1 + n_2. \tag{1}$$

It is well known in topology [2] that this quantity is independent of the triangulation chosen. In Fig. 3, the image has $E=1$ in both cases. Any simply connected object has $E=1$.

Clearly if the image consists of B simply connected objects, $n_0-n_1+n_2$ counted over the entire image would be just B. Next, suppose that a hole is introduced into the image, for example, by drawing a line through the 0's area between two existing vertices on a border. This change gives $\Delta E=0-1+0=-1$. Clearly $\Delta E=-1$ will hold for any distinct hole added to the image. Then if the complete image consists of B objects and L holes (distributed arbitrarily among the objects), we will have

$$E = B - L = n_0 - n_1 + n_2. \tag{2}$$

Triangulation implies that Euler number is locally countable because all we need to do is subdivide the image into sufficiently small cells, and count n_0, n_1, and n_2 in each cell, summing at the end. This may be counterintuitive since the image can have arbitrarily large and intertwined objects and holes. (It has been proven that neither B nor L can be separately determined by local property measurements [1, p. 134].

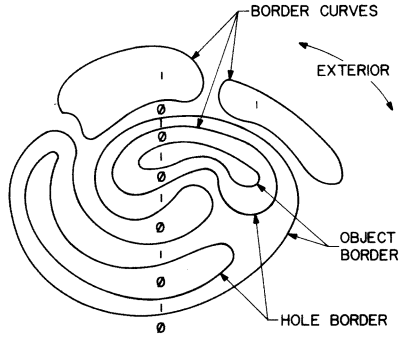


Fig. 2. Borders, objects, holes in two dimensions.

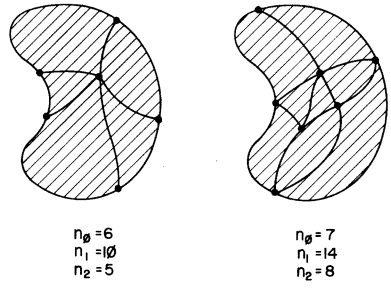


Fig. 3. Two triangulations of one object.

"Triangulation" need not be into areas with three sides—any number greater than zero is valid as long as all areas and border line segments are simply connected. A line may have both ends terminating at the same point. For images containing lower dimensional objects, triangulation may include points and lines as objects, as well as areas.

C. Image Combining

We wish to show that Euler number (E) is an "additive set property"—that is, given two images I_1 and I_2 and their Euler numbers $E(I_1)$ and $E(I_2)$, the Euler numbers of the union and intersection images are constrained by

$$E(I_1) + E(I_2) = E(I_1 \cap I_2) + E(I_1 \cup I_2). \tag{3}$$

Similarly, we show that for 1's area A,

$$A(I_1) + A(I_2) = A(I_1 \cap I_2) + A(I_1 \cup I_2) \tag{4}$$

and for border length (perimeter) P,

$$P(I_1) + P(I_2) = P(I_1 \cap I_2) + P(I_1 \cup I_2). \tag{5}$$

First, for (3) suppose we generate a triangulation which is valid for I_1, I_2, $I_1 \cap I_2$, and $I_1 \cup I_2$, in the sense of Fig. 4. That is, a set of points $\{a, \cdots, g\}$ and lines is chosen such that this set or a subset of it properly triangulates all four images. First, considering the points, some will belong to both I_1 and I_2 (b, c, f, g). These will belong to both $I_1 \cap I_2$ and $I_1 \cup I_2$. Others will belong to I_1 or I_2 but not both (a, d, e). These will belong to $I_1 \cup I_2$ but not $I_1 \cap I_2$. It should then be clear that

$$n_0(I_1) + n_0(I_2) = n_0(I_1 \cap I_2) + n_0(I_1 \cup I_2).$$

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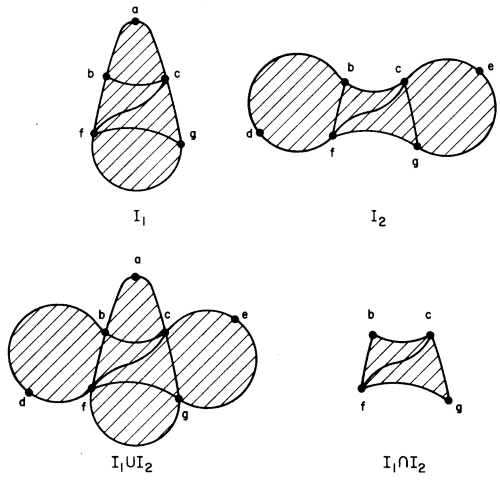


Fig. 4. Triangulation of images and combinations.

(In Fig. 4, this equality represents $5+6=7+4$.) A similar argument will demonstrate the above for n_1. The equality for the figure is $8+9=12+5$; similarly for n_2, for which the example gives $4+4=6+2$. From these equalities and (2), (3) immediately follows.

In the argument for n_2, the totality of actual subareas is identical for the two sets of images, as well as their number. From this, (4) follows. Next, note that any part of the perimeter of I_1 "outside" I_2 will be perimeter in $I_1 \cup I_2$, but not in $I_1 \cap I_2$. This and similar reasoning establishes (5).

(For (5) to hold for two objects lying on opposite sides of a common border, in which case $I_1 \cap I_2$ contains lines which are not borders of objects, the perimeter of such a line is considered to be twice its length.)

We now examine several consequences of the additive set property which will be useful later on.

1) Suppose we have an initial image I, to which we add (in the sense of union) an additional piece ΔI. Each of the additive set properties, for example E, undergoes a change ΔE when ΔI is added. From (3) we have

$$\Delta E \equiv E(I \cup \Delta I) - E(I) = E(\Delta I) - E(I \cap \Delta I) \qquad (6)$$

that is, the change in E is equal to E of the added piece minus E of the intersection. Similarly for P or A.

2) From (3), using the associative and distributive laws for sets, we can prove (by induction on N) a law for combining N images:

$$E(I_1 \cup I_2 \cup \cdots \cup I_N) = \sum_{i=1}^{N} E(I_i) - \sum_{i=1}^{N} \sum_{j=i+1}^{N} E(I_i \cap I_j) + \sum_{i=1}^{N} \sum_{j=i+1}^{N} \sum_{k=j+1}^{N} E(I_i \cap I_j \cap I_k) - , + , -, \text{etc.,}$$

to the last term, $+(-1)^{N-1} E(I_1 \cap I_2 \cap \cdots \cap I_N)$. (7)

Example: Suppose a binary image consists of N overlapping rings, such that each ring meets every other ring in two places, with no three rings meeting in a point. How many holes does the image contain? We use (7), considering each ring to be one object with one hole, therefore having

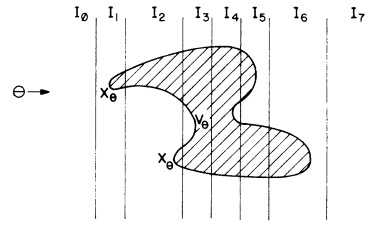


Fig. 5. Critical points in two dimensions.

$E=0$. The combined image has $B=1$, being all joined together. Each term in the first sum in (7) is 0. Each term in the double sum is 2, and there are $N(N-1)/2$ terms. All further sums are 0. Hence $1-L(N)=-2N(N-1)/2$, so $L(N)=N^2-N+1$.

3) Consider a base image I to which we join two other images, A and B. Each image added separately produces, respectively, ΔE_A and ΔE_B. Under what conditions do the separate changes sum to give the change produced by joining $A \cup B$? A simple derivation based on (6) gives

$$\Delta E_{A \cup B} - \Delta E_A - \Delta E_B = E(I \cap A \cap B) - E(A \cap B) \qquad (8)$$

which yields the following lemma.

Lemma: If A and B are disjoint images, the two joined simultaneously to an image I produces the same ΔE as if A and B are joined one after another.

If $E(I \cap A \cap B) \neq E(A \cap B)$, the union of A and B onto I is "nonlinear." Note that the union can be linear even if $A \cap B \neq 0$.

D. Critical Point Approach

Suppose an image I is split into a sequence of subimages $I_1, I_2, \cdots, I_N$ by a series of nonintersecting lines, so that subimage I_i touches only I_{i-1} and I_{i+1}. For simplicity, let the dividing lines be more or less parallel, as in Fig. 5, and let θ be a "vector" pointing downstream, that is, in the direction of increasing index (i).

Consider a subimage, such as I_1 or I_2, which contains one upstream-facing convexity (denoted X_θ) but no upstream-facing concavities (such as V_θ in I_3). Now a convexity constitutes one object in its subimage I_i; but since the convexity, by definition of the subimages, does not touch the border between I_i and I_{i-1}, the one-dimensional image $I_i \cap I_{i-1}$ does not have a corresponding object. Assuming that all other objects in I_i extend back to the line $I_i \cap I_{i-1}$, it is then true that

$$E(I_i) - E(I_i \cap I_{i-1}) = 1.$$

Now consider a subimage I_i containing one V_θ and no X_θ's. The V_θ is associated with one object in I_i; this object touches $I_i \cap I_{i-1}$ in two places. Therefore,

$$E(I_i) - E(I_i \cap I_{i-1}) = -1.$$

Note that the image also contains points $X_{-\theta}$ and $V_{-\theta}$, which are not being counted, because for such a point,

$$E(I_i) = E(I_i \cap I_{i-1}).$$

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If we now let $n(X_\theta)$ and $n(V_\theta)$ denote the number of left-facing convexities and concavities, respectively, in the image, we will have

$$\sum_{i=1}^{N} E(I_i) - \sum_{i=1}^{N} E(I_i \cap I_{i-1}) = n(X_\theta) - n(V_\theta). \quad (9)$$

Now refer to (7). All terms on its right for the present case are zero except the first sum and the members of the second sum involving intersections of adjacent subimages. The double sum in (7) thus reduces to the second sum in (9). Hence we have

$$E(I) = B - L = n(X_\theta) - n(V_\theta). \quad (10)$$

This equation yields extremely simple formulas for Euler number on various lattices. In the above treatment, X_θ and V_θ do not denote specific points on the object (or hole) borders, but rather small regions, whose location is relative to the chosen family of separating lines. The term "critical point" is taken from topology.[3]

An interesting fact about Euler number, triangulation, and critical points is that the overall approach extends readily to three dimensions [4], which will be covered in a subsequent paper.

E. Image Characterization; Applications

Certain formulas involving the basic local properties E, P, and A provide useful geometric characterizations of binary images. In the following analysis let us assume that the image contains very few holes compared to the number of objects, so that $E \cong B$. First, average area $\bar{A}$ and perimeter $\bar{P}$ per object are given by

$$\bar{A} = A/E$$
$$\bar{P} = P/E \quad (11)$$

which would be useful in distinguishing speck-type images from blobby or lumpy ones.

Further measures are available for images known to consist of long thin line-type objects. Such images include printing, writing, line drawing, and certain micrographs of fibrous structures. Line length k, average line width w, and the average length/width ratio k/w are given by

$$k = P/2 \quad (12)$$
$$w = A/k = 2A/P \quad (13)$$
$$k/w = P^2/4A. \quad (14)$$

On the other hand, if the image consists of a circular object of radius r, $P = 2\pi r$ and $A = \pi r^2$, so $P^2/4A = \pi$. The constant π is the minimum value for $P^2/4A$, attained for any image consisting of one or more circular disks. Therefore, $P^2/4A$ is a measure of the average elongation of objects in an arbitrary image. (The quantity P^2/A has been used by psychologists to define the apparent "dispersion" of a picture [11].)

These measures, which are very simple to compute, are extremely useful in practice. A particularly good example is for evaluating a binary image of a printed character. The average line width w should be 0.1 to 0.2 of the character width for clean typing [5]. Also, k/w should be at least 10 for high-quality print. E itself should ideally be between -1 (for 8 or B) and $+2$ (for i or j). (E is very noise sensitive, so must be used with caution.) If these tests fail, an improved image should be obtained, rather than entering a lengthy recognition process.

[3] See Bott [3] for a technical presentation of critical point theory.

III. Cellular Binary Images

A regular tessellation of the plane, denoted (p, q), consists of (q) regular p-gons coming together at a point, with every p-gon having p such points. Thus the p-gons and points cover the plane. It is easily proven [6] that the only three such lattices are (6, 3), (4, 4), and (3, 6), shown in Fig. 6. We consider the p-gon to be a "bit cell" of a cellularized image. If the centers of adjacent bit cells are connected with lines, we get q-gons, called "cell groups."

Having these definitions, we can be precise about what a sufficiently fine image cellularization is. Assume that any cellular binary image is a quantization of an original binary image having smooth border curves. Let us first assume that the radius of curvature of all border curves is everywhere greater than the diameter of a bit cell, or equivalently, than of a celi group. Suppose the binary decision for a given bit cell is based on whether the cell is of value 1 over more or less than 50 percent of its area. This is approximately equivalent, for low-curvature borders, to whether the border curve includes the bit cell center point in the 0 region or in the 1 region.

Since bit cell centers define the cell group, we investigate how border curves can penetrate these polygons. One way to define a sufficiently fine cellularization is: the bit cell dimension must be smaller than the smallest significant distance in the image. This implies that no bit cell or cell group polygon contains more than one entry or exit of a border curve.

Definition: A cell group polygon in a particular image is *separable* if and only if a straight line can be drawn through it which has all the 1's on one side and all the 0's on the other side. A *separable image* is a cellularized image having no inseparable cell groups. Then a *sufficiently resolved cellularization* is one which generates a separable image from the original. For the (6, 3) cell group, clearly any possible cell group can result from one border entering and exiting the triangle. Therefore, for the (6, 3) hexagonal lattice, all cell groups are separable, hence the well-known fact [7], [8] that the hexagonal lattice has no diagonal connection complication.

For the square lattice, there are six distinct types of cell groups, of which $\left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$ is obviously inseparable. Alternatively, two of the 16 total types are inseparable. For the (3, 6) lattice of triangles, there are 13 distinct types and 64 total types of cell groups. Six distinct types and 32 total types are inseparable. Clearly, the inseparability fraction rises with q in (p, q).

The "neighborhood" $\boldsymbol{U}$ of a bit cell U is the set of all bit cells whose state must be known to determine what effect

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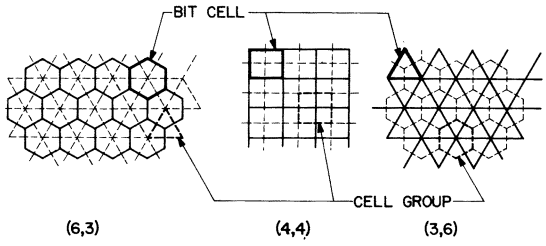


Fig. 6. Regular plane tessellations or lattices.

changing the value of U has on the Euler number. As will be seen, for the hex lattice, $\boldsymbol{U}$ consists of the six adjacent hexagons; for the square lattice, $\boldsymbol{U}$ is the four edge-adjacent and four vertex-adjacent squares. For the (3, 6) lattice, $\boldsymbol{U}$ is the three edge-adjacent and nine vertex-adjacent triangles. Clearly the complexity of a given lattice is directly related to the complexity of its cell group polygon. Because of this, and because it has fewer equivalent radial directions (3) than the others, the (3, 6) lattice does not seem to have any advantages for image processing, and is not discussed in detail.

IV. The Hexagonal Lattice

The (6, 3) tessellation is examined first, because its neighborhood connectivity properties are much simpler than those of (4, 4).

A. Euler Number Algorithm

An explicit formula for E on the hexagonal lattice is most easily derived using (10). The proper method of subdividing the image is to choose the stream direction θ parallel to a side of the bit cell hexagons. See Fig. 7. Clearly there is only one pattern of three bits which can constitute an X_θ, and one other which can make a V_θ. With this choice of θ, we can write

$$n(X_\theta) = n\begin{bmatrix} 0 & 0 \\ & 1 \end{bmatrix}$$
$$n(V_\theta) = n\begin{bmatrix} & 0 \\ 1 & 1 \end{bmatrix}. \tag{15}$$

Then from (10) we immediately have

$$E = n\begin{bmatrix} 0 & 0 \\ & 1 \end{bmatrix} - n\begin{bmatrix} & 0 \\ 1 & 1 \end{bmatrix}. \tag{16}$$

Since there are six equivalent choices of θ, there are six equivalent forms of (16), which yield exactly the same value for E if the image has a zero exterior. A direction-symmetrical form of (16) can be derived by defining T_1 as the number of cell group triangles in the image having one 1 and two 0's, in any of the six orientations, and similarly for T_2. By adding the six versions of (16) we get

$$E = (T_1 - T_2)/6. \tag{17}$$

B. Perimeter Length

To obtain a simple formula for perimeter length P, consider a straight horizontal line bisecting vertical hexagon sides. Any such line entering a succession of 1's must leave

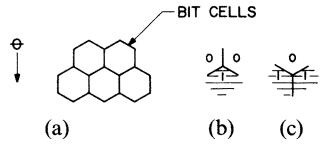


Fig. 7. X_θ and V_θ on hexagons. (a) Choice of θ. (b) X_θ. (c) V_θ.

it. This argument gives, for any image having a zero exterior,

$$n[01] = n[10]$$
$$n\begin{bmatrix} 0 \\ 1 \end{bmatrix} = n\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$n\begin{bmatrix} 0 \\ 1 \end{bmatrix} = n\begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{18}$$

Now clearly the perimeter length, in terms of hexagon side lengths, is equal to the number of unit edges of all orientations. The six orientations can be combined by (18) into three sets, so

$$P = 2n[01] + 2n\begin{bmatrix} 0 \\ 1 \end{bmatrix} + 2n\begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{19}$$

Since by (18) each term has an equivalent, there are eight ways to write (19).

P can also be found by summing over cell group triangles. Any T_1 or T_2 contains one unit of perimeter length, so

$$P = T_1 + T_2. \tag{20}$$

Equation (20) will always be an overestimate for originally smooth straight perimeters. An approximate perimeter P_a, having zero error averaged over all perimeter angles, is given by

$$P_a = (T_1 + T_2)/1.24. \tag{21}$$

C. Euler Differential; Neighborhood Processing

Consider a given hexagonal bit cell U in an arbitrary image. If this cell is 0, the image Euler number is denoted E_0; setting this cell to 1 gives a different Euler number, E_1. Note that if $\Delta E_1 = E_1 - E_0$ is the change in E due to setting U to 1, then $\Delta E_0 = E_0 - E_1 = -\Delta E_1$ would be the change if U is reset. Denoting the bit cell change as ΔU ($=1$ for setting, -1 for resetting), and the resulting Euler number change as ΔE, clearly the ratio $\Delta E/\Delta U$ is the same whether $\Delta U = +1$ or -1. We then define the Euler differential $E^*(\boldsymbol{U})$ of a particular neighborhood $\boldsymbol{U}$ as

$$E^*(\boldsymbol{U}) = \Delta E/\Delta U. \tag{22}$$

Since Euler number is locally countable, clearly the sums needed to find E_0 and E_1 will differ only over the six cell group triangles which contain the changing bit U. Therefore $E^*(\boldsymbol{U})$ must be a function of the immediate six-bit neighborhood of U, denoted $\boldsymbol{U} = \{U_0, \cdots, U_5\}$, where the U_i are numbered sequentially clockwise (and subscripts are treated modulo 6).

To find E^*, we calculate the changes in n_0, n_1, and n_2 over the six-bit cell neighborhood resulting from setting U from 0 to 1, and use $\Delta E = \Delta n_0 - \Delta n_1 + \Delta n_2$. To do so, we adopt

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an image "triangulation" convention where every "1" bit cell has or shares six edges contributing to n_1, and six vertices contributing to n_0, regardless of how many other "1" cells the given cell touches. (That is, the interior of all objects is completely "hexagonalized.") First, setting U from 0 to 1 always causes $\Delta n_2 = +1$. Secondly, with the convention, Δn_1 will be $+1$ for every "0" bit cell adjacent to U. Last, Δn_0 will be $+1$ for every pair of consecutive zeros in $\boldsymbol{U}$. Therefore the following lemma holds.

Lemma: $E^*(\boldsymbol{U})$ is equal to the number of 0, 0 adjacent cell pairs in $\boldsymbol{U}$, minus the number of 0 cells in $\boldsymbol{U}$, plus one.

Now since each 0 cell must have a 1 or a 0 as its clockwise neighbor, we can write $n(0)=n(0,\ 1)+n(0,\ 0)$. Then $E^*=n(0,\ 0)-n(0)+1=1-n(0,\ 1)$. For purpose of BIP design, we wish to express E^* in terms of the population or 1's count of a (possibly transformed) neighborhood of U. If we define a new neighborhood $\boldsymbol{U}'$ in terms of

$$U_i' = U_i\overline{U}_{i-1} \tag{23}$$

clearly the number of 1's in $\boldsymbol{U}'$, denoted $|\boldsymbol{U}'|$, will be just $n(0,\ 1)$ of the original U. Then, finally,

$$E^*(\boldsymbol{U}) = 1 - |\boldsymbol{U}'|. \tag{24}$$

For examples, see Fig. 8.

Fig. 8(a) is typical of a bit lying on the edge of an object. Fig. 8(b) shows a bit which if set will join two objects, or perhaps close off a hole. In Fig. 8(c), the center bit joins three branches, for example of a skeleton image. In Fig. 8(d), $E^*=+1$, because setting the center bit creates a new object. In Fig. 8(e), setting the center bit removes a hole, so $E^*=+1$.

Resetting bits whose $E^*=0$ is the basic operation needed to skeletonize a hexagonal image, since a skeleton by definition is the thinnest image having the same connectivity as the original. However, all such bits cannot be removed at once, because of the possibility of a nonlinear interaction. The lemma of Section II-C says that if two bits A and B, each having $E^*=0$, are reset, the total E^* is guaranteed to be 0 if A and B are disjoint. (Equation (8) is written in terms of the union of new images, so we can think of adding bits to the complement image.) Since the hexagonal lattice can be separated into three congruent regular disjoint sets, a correct skeletonizing procedure consists of groups of three steps. This is discussed in more detail by Golay [8, p. 734].

Since P and A are locally summable just as E is, the six-bit neighborhood $\boldsymbol{U}$ is also associated with a differential area and differential perimeter, defined according to

$$A^*(\boldsymbol{U}) = \Delta A/\Delta U = 1$$

and

$$P^*(\boldsymbol{U}) = \Delta P/\Delta U. \tag{25}$$

To find P^*, we use (6) in the form

$$\Delta P = P(\Delta I) - P(I \cap \Delta I) \tag{26}$$

where I is the center hex cell. Then $P(I\cap\Delta I)=2|\boldsymbol{U}|$, because a line segment in an image has a perimeter of twice its length, and a unit line segment results in the intersection wherever the neighborhood was 1. Since $P(\Delta I)=6$, we have

$$P^*=6-2|\boldsymbol{U}|. \tag{27}$$

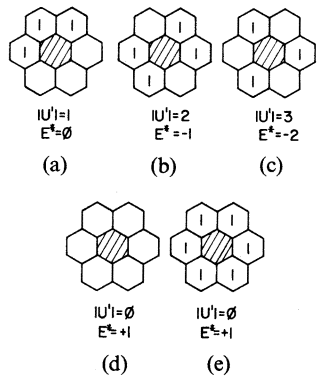


Fig. 8. Examples of Euler differential.

V. The Square Lattice

A. Bit Quads

We have introduced bit cells, which are squares each having the value 0 or 1, and cell groups, which are squares centered on the corners of four bit cells. A *bit quad* is a 2×2 array of bit cells. Clearly the values of the four bit cells in a quad must be known to count the fundamental quantities n_0, n_1, and n_2 in the corresponding cell group.

There are 16 possible bit quads, and six distinct types considering rotational equivalence: Q_i, $0\le i\le 4$, contains (i) 1's; Q_D, contains the types $\left[\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right]$ and $\left[\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right]$. The notation $n(Q_i)$ or $n(Q_D)$ refers to the number of the corresponding type in the image. It is not difficult to prove the following formulas involving image area A and perimeter P.

$$4A = n(Q_1) + 2n(Q_2) + 2n(Q_D) + 3n(Q_3) + 4n(Q_4) \tag{28}$$

$$P = n(Q_1) + n(Q_2) + 2n(Q_D) + n(Q_3) \tag{29}$$

$$P \text{ and } n(Q_2) \text{ are even integers.} \tag{30}$$

B. Perimeter Length; Approximate Area

An argument similar to that used with hexagons gives

$$P = 2n[01] + 2n\begin{bmatrix}0\\1\end{bmatrix} \tag{31}$$

which is exactly equivalent to (29). This formula can err up to 41 percent if we are determining the length of originally smooth (uncellularized) perimeters. For such purposes, an approximation accurate to 8 percent can be obtained in which we assume a Q_1 or Q_3 bit cell represents a 45° "shortcut" taken by the original smooth contour. This approximate perimeter P_a is given by

$$P_a = n(Q_2) + [n(Q_1) + n(Q_3)]/\sqrt{2}. \tag{32}$$

Similar reasoning yields an approximate area A_a:

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$$4A_a = n(Q_1)/2 + n(Q_2) + n(Q_D) + 7n(Q_3)/2 + 4n(Q_4). \quad (33)$$

(For images with low Euler number, (33) and (28) give almost identical results.)

C. Euler Number

The topology of the square lattice is obviously complicated by the Q_D quad, which can be treated in several ways: vertex-adjacent 1-cells in a Q_D can be considered connected; the 0-cells in a Q_D can be considered connected; or some assumption which is symmetrical with respect to 0 and 1 can be made. We will show below that these and other answers are mathematically consistent.

To derive the basic formula for E, we adopt a "triangulation" convention in which all 1's areas are divided into unit squares by adding suitable vertices and line segments, just as in the hexagonal case. See Fig. 9. We now count n_0, n_1, and n_2, summing over the cell group squares. Allowing for the fact that a cell group square can contain only half of any unit line segment, and only one-quarter of any bit cell, we see that square 1 contains one vertex, two halves of line segments, and 1/4 of a subarea. Clearly, the contribution of a cell group depends only on its quad type, and in detail are as follows.

Square	Quad Type	Δn_0	Δn_1	Δn_2	$\Delta(n_0 - n_1 + n_2)$
0	Q_0	0	0	0	0
1	Q_1	1	1	1/4	1/4
2	Q_2	1	3/2	1/2	0
3	Q_3	1	2	3/4	−1/4
4	Q_4	1	2	1	0
5	Q_D	1	2	1/2	−1/2

Summing over all cell groups gives counts over the entire image; thus,

$$4\sum \Delta(n_0 - n_1 + n_2) = n(Q_1) - n(Q_3) - 2n(Q_D).$$

But summing $\Delta(n_0 - n_1 + n_2)$ over the image gives the Euler number, denoted W:

$$4W = n(Q_1) - n(Q_3) - 2n(Q_D). \quad (34)$$

In the above treatment, a diagonal connection such as $\left[\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right]$ is considered to be a full connection, since the two bit cells meet (in one point). Hence the symbol W, standing for "ones-joined Euler number". Clearly there should be a formula which treats two diagonally adjacent 1's to be disconnected. This is derived by imagining the two 1's cells corners not touching, thus requiring two vertices; hence for a Q_D,

$$\Delta(n_0 - n_1 + n_2) = +1/2.$$

This Euler number will be denoted Z ("zeros joined"), and is given by

$$4Z = n(Q_1) - n(Q_3) + 2nQ_D). \quad (35)$$

A "half-joined" Euler number can be defined as the average

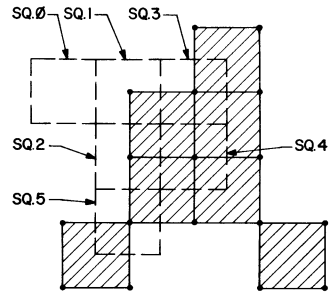


Fig. 9. "Triangulation" of square lattice image.

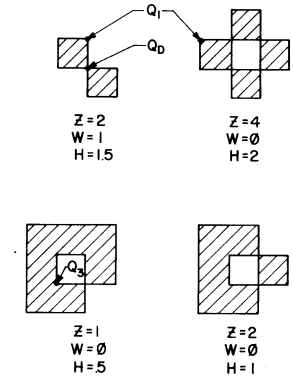


Fig. 10. Examples of three Euler numbers.

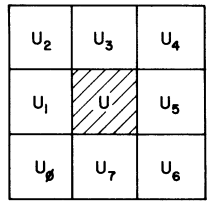


Fig. 11. Square lattice neighborhood nomenclature.

of W and Z:

$$4H = n(Q_1) - n(Q_3). \quad (36)$$

The H model of the plane considers a Q_D to be a half-connection. See Fig. 10.

D. Eight-Bit Neighborhoods and Euler Differential

A given bit cell is a member of four quads, the union of which is a 3×3 array of bits. Notation is shown in Fig. 11. Subscripts are taken modulo 8. The center cell U is not considered part of the eight-bit neighborhood $\boldsymbol{U}$. (There are 51 distinct types of neighborhoods, allowing for rotational and reflectional equivalence.) Since A, P, H, W, and Z are all locally countable, a given eight-bit neighborhood is associated with the differentials $A^*(\boldsymbol{U}), \cdots, Z^*(\boldsymbol{U})$, defined exactly as in Section IV-C. To find a formula for $P^*(\boldsymbol{U})$, we follow the argument used for (27), noting that only the neighborhood of odd-numbered cells, denoted $\boldsymbol{U}_d = \{U_1, U_3, U_5, U_7\}$, contributes to $P(I \cap \Delta I)$. [See (27).] In this case $P(\Delta I) = 4$, so we have

$$P^* = 4 - 2|\boldsymbol{U}_d|. \quad (37)$$

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To find W^*, Z^*, and H^*, we calculate the changes in n_0, n_1, and n_2 over the neighborhood resulting from changing U from 0 to 1. We again assume complete subdivision of 1's areas into points, lines, and unit subareas. Since setting U always adds a subarea, obviously $\Delta n_2 = 1$. Next, for each zero bit cell edge-adjacent to U, a unit line is added. Finally, a vertex will be added at a particular corner of U only if there is no vertex there when $U=0$. This can only be if the three adjacent bits in $\boldsymbol{U}$ are all 0. Therefore the following lemma holds.

Lemma: W^* is equal to the number of $\{0, 0, 0\}$ corners minus the number of zeros among the odd neighbors, plus one.

Considering the possible values of a given corner bit and its two neighbors, such as the corner $\{U_3, U_4, U_5\}$, only the configurations $\{0, 1, 0\}$, $\{1, 0, 0\}$, and $\{1, 1, 0\}$ contribute to W^*. Denoting the number of such three-bit combinations taken clockwise, as C_{246} (range 0 to 4), we have

$$W^*(\boldsymbol{U}) = 1 - C_{246}. \tag{38}$$

A corresponding formula can be derived for $Z^*(\boldsymbol{U})$ by considering image complements. We get

$$Z^*(\boldsymbol{U}) = 1 - C_{135} \tag{39}$$

where C_{135} is the number of corner-centered bit-triplets having any of the patterns (taken clockwise) $\{0, 0, 1\}$, $\{0, 1, 1\}$, or $\{1, 0, 1\}$.

Finally, defining C_{07} as the number of "homogeneous" corners, we have

$$H^*(\boldsymbol{U}) = C_{07}/2 - 1. \tag{40}$$

See Fig. 12.

It is then possible to define a simple Boolean transformation of $\boldsymbol{U}$ into $\boldsymbol{U}'$, such that the populations of $\boldsymbol{U}'$ and its subsets give

$$W^*(\boldsymbol{U}) = 1 - |\boldsymbol{U}'_e| \tag{41}$$

$$Z^*(\boldsymbol{U}) = 1 - |\boldsymbol{U}'_d| \tag{42}$$

$$H^*(\boldsymbol{U}) = 1 - |\boldsymbol{U}'|/2. \tag{43}$$

The transformations and detailed proofs of (38)–(40) are given in [10].

E. Skeleton Extraction

As an example of the above algorithms, consider having an arbitrary binary image on the square lattice, perhaps consisting of line-like elements. From this image we can generate a "weakly connected" skeleton; that is a unit-width line image, topologically equivalent to the original image, in which a Q_D is a valid join in the skeleton. This can be done by resetting each U whose $W^*(\boldsymbol{U})=0$. As explained for the hexagonal lattice, only disjoint points may be removed in any one step. The square lattice must be considered as the union of four (or more) congruent regular disjoint sets. Therefore the skeletonizing procedure consists of sets of four substeps. A "strong" skeleton may be generated by removing points whose $Z^*(\boldsymbol{U})=0$, using the same disjoint sets for simultaneous removal.[4] (It happens that it is almost always satisfactory, for obtaining the weak skeleton, to allow simultaneous removal over half the lattice, rather than one-quarter. This makes weak skeletons easier to extract.)

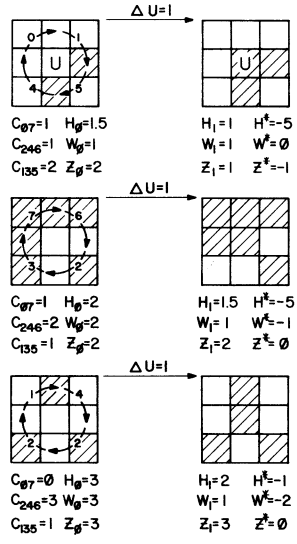


Fig. 12. Examples of square lattice Euler differentials.

Another approach to skeletonizing is sequential, in which a cell whose $E^*=0$ is reset, and then its new value of 0 is used in computing the neighborhood of the next cell. This will always yield a topologically valid skeleton, but one whose metric properties, for example line position with respect to the original areas, may be undesirable.

The above discussion of skeletonizing omits certain post-processing cleanup steps which are desirable to smooth the skeleton. Also not discussed is stabilization of end points, which is simple, but is best described in the context of the binary image processor hardware. This application and others will be covered in a future paper.

F. Critical Point Euler Number

The Euler numbers W, Z, and H were all symmetrical functions of the quads, that is, the directions right and left are treated identically, as are up and down, and as are the two types of Q_D. Critical points give simpler formulas which, however, are direction dependent. If θ is chosen to point "southeast," clearly an X_θ can only arise from $\left[\begin{smallmatrix}00\\01\end{smallmatrix}\right]$, and a V_θ must be $\left[\begin{smallmatrix}01\\11\end{smallmatrix}\right]$. Then (11) gives

$$E = n\begin{bmatrix}00\\01\end{bmatrix} - n\begin{bmatrix}01\\11\end{bmatrix}. \tag{44}$$

If we assume that Q_D's are rare, a bit can be dropped from each [ ], because its value is implied, giving

[4] The above argument carries over directly to the triangular lattice. The formula for $W^*(\boldsymbol{U})$ is: the number of $\{0, 0, 0, 0, 0\}$ corners, minus the number of zeros among the edge-adjacent cells, plus one. To skeletonize, remove cells whose $W^*=0$; this must be done over disjoint sets of cells. The triangular lattice is the union of six congruent regular sets, making skeletonizing on this lattice relatively slow.

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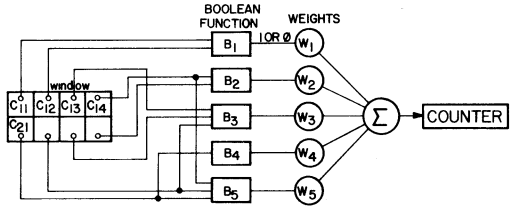


Fig. 13. Typical LPC. Image dimension $N=2$. LPC diameter $D=4$. LPC degree $G=5$. LPC order $R=3$ (because of B3, B5). LPC dimension $M=2$ (because of B3).

$$E = n\begin{bmatrix} 0 \\ 01 \end{bmatrix} - n\begin{bmatrix} 01 \\ 1 \end{bmatrix} \tag{45}$$

and three equivalent expressions. There are also eight other expressions for E, easily derived from (44), of the form

$$E = n\begin{bmatrix} 00 \\ 1 \end{bmatrix} - n\begin{bmatrix} 0 \\ 11 \end{bmatrix}. \tag{46}$$

The three formulas above will treat one type of Q_D as a ones-join, and the other as a zeros-join. This direction dependence can be "averaged out":

$$2H = n\begin{bmatrix} 0 \\ 10 \end{bmatrix} + n\begin{bmatrix} 0 \\ 01 \end{bmatrix} - n\begin{bmatrix} 10 \\ 1 \end{bmatrix} - n\begin{bmatrix} 01 \\ 1 \end{bmatrix}. \tag{47}$$

We have presented four valid ways of treating the diagonal connection Q_D. In addition, one can consider $\left[\begin{smallmatrix} 10 \\ 01 \end{smallmatrix}\right]$ to be ones-joined on even-numbered rows and zeros-joined on odd-numbered rows; $\left[\begin{smallmatrix} 01 \\ 10 \end{smallmatrix}\right]$ is treated exactly opposite. This would obtain naturally if we moved the even rows 1/2 cell to the left, which figure resembles the hexagonal lattice. Since all these treatments are valid, Q_D should certainly be regarded as a complication of the square lattice, but not as a paradox or ambiguity. It is not a unique complication, but a result of the fact that the neighborhood complexity of the (p, q) tessellation increases with q.

For a related but quite different treatment of topology on the square lattice, refer to Rosenfeld [13].

VI. The Local Property Counter

A. Definitions

This section discusses image measurements which can be made by a particular type of perceptron-like device, called the local property counter (LPC), operating on discrete binary images in one and two (and later, three) dimensions. Rather than describing the LPC in terms of standard perceptron notation, we choose a serial analog, as follows. (Assume a two-dimensional image for now.)

A binary image is defined on the two-dimensional finite square lattice in the usual way, and is assumed to be surrounded by zeros. A rectangular window of IJ cells, U_{ij}, $i=1, \cdots, I, j=1, \cdots, J$, is simultaneously sampled by a digital computing device, which calculates some fixed Boolean function F on the IJ bits. If the Boolean function is true, a counter is incremented. We actually allow several Boolean functions $F_1, F_2, \cdots, F_G$ to be simultaneously computed, all affecting the same counter. See Fig. 13.

Associated with each function F_g is a weight W_g. The window serially takes every possible position in the image, for example by scanning in a raster fashion. When the scan is over, the counter contains a "locally countable property" of the image.

Definitions (for an N-dimensional image) are as follows.

Symbol	Name	Definition
D	Diameter[5]	Max (I, J)
G	Degree	Number of F_g's feeding the counter[6]
R	Order[5]	Maximum over g of the number of cells in the window feeding any F_g
M	Dimension	Maximum over g of the number of orthogonal axes "spanned" by the sample points of any one function F_g

Example 1: An LPC for measuring the area A (number of 1's) of a two-dimensional binary image. This obviously requires a 1×1 window, hence $D=1$ and $M=0$. We need one Boolean function feeding the counter, namely $F_1 = U_{11}$. Hence $G=1$ and $R=1$. This computation can be represented by $A=n[1]$.

Example 2: An LPC for measuring the perimeter length P of a two-dimensional binary image. This LPC calculates $P=2n[01]+2n\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$. This LPC has diameter $D=2$, degree $G=2$, order $R=2$, dimension $M=1$.

The local property counter is analogous to a perceptron each of whose second-level elements computes a weighted sum of all its first-level outputs, and compares the sum to a threshold, producing a final decision of 1 or 0. An LPC is primarily intended to measure or count properties, not for deciding or recognizing, so the threshold step is omitted.

Having defined the LPC, we now define the *localness* of a measurement on a binary lattice as the ordered set of integers $\{D, G, R, M\}$, which are the minimum diameter, degree, order, and dimension on LPC needs to determine the measurement.

[5] These terms were chosen to correspond with typical restrictions placed on the first-level elements in a parallel perceptron [1].

[6] The split into (G) functions is somewhat arbitrary, but the F_g's have been chosen so that G is an indicator of the complexity of the counting logic.

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B. Summary of Local Properties

Image Dimension	Property	Symbol	See Equation	LPC Characteristics[7] Diameter D	Degree G	Order R	Dimension M
1	Length (total 1's)	L	—	1	1	1	0
2	Area	A	—	1	1	1	0
2	Approximate area (square)	A_a	(33)	2	5	4	2
2	Perimeter length	P					
	hex		(19)	2	3	2	1
	square		(31)	2	2	2	1
2	Approximate perimeter	P_a					
	hex		(21)	2	3	2	1
	square		(32)	2	3	4	2
1	Euler number	E	—	2	1	2	1
2	Euler number	E	(17)	2	2	3	2
	critical point		(46)	2	2	3	2
2	Euler number, direction independent	W	(34)	2	3	4	2
		Z	(35)	2	3	4	2
		H	(36)	2	2	4	2
		H	(47)	2	4	3	2

C. Remarks on Localness

The table shows that on an N-dimensional lattice, Euler number E is a measurement of order $N+1$ and diameter 2, at least for $N \leq 3$. (It has been shown that E is of order 4 on the cubic lattice [4, p. 54].) The quantity $N+1$ is what one might expect, for $N+1$ points are required to define N orthogonal directions. It is interesting to compare this with a previous result given for two dimensions [1, p. 87]:

$$E = n[1] - n[11] - n\begin{bmatrix} 1 \\ 1 \end{bmatrix} + n\begin{bmatrix} 11 \\ 11 \end{bmatrix}. \tag{48}$$

(This expression can easily be derived from the $E=n_0 - n_1 + n_2$ argument.) Their expression easily generalizes to three (and more) dimensions. There would be one "point term," $n[1]$, three linear terms, three area terms, and a new volume term, namely $-n[2\times 2\times 2$ cubes of 1's]. Hence, E in N dimensions is clearly of order $R \leq 2^N$ with $D=2$. I do not know of a proof that E is of order exactly $N+1$.

All local properties considered in this paper are of diameter 2 or less. Since these properties and the derived characterizations (width, thickness, etc.) are fairly powerful measurements, and since no useful local properties of higher diameter are known, it is suggested that diameter-2 neighborhoods are sufficient for powerful numerical image measurements. In Sections IV-C and V-D, we found that diameter-3 neighborhoods are necessary for finding certain cells in the image (e.g., those having a particular Euler differential, in terms of $|U'|$, or those having a certain neighborhood population $|U|$). These contrasting principles are not put forward as being absolute; for example, it is clear that larger diameter measurements can yield a more accurate expression for P_a (assuming originally smooth boundaries); further, certain diameter-3 cell-finding operations could be done by a series of diameter-2 operations.

We also note that certain properties change order and degree with the lattice type, but keep the same diameter, for example P and P_a. Diameter is also constant with image dimension for certain measures, such as E. This strongly suggests that diameter D is the most fundamental localness parameter.

Lastly, it is interesting that in two dimensions, H may be a slightly more fundamental measure than Z or W, since it can be obtained by either a smaller order or a smaller degree expression.

VII. The Binary Image Processor (BIP)

We have shown that some important image measurements are locally countable during a sequential pass through the picture. Having discovered these properties, we were unwilling to discard them, so we considered the feasibility of an entirely serial binary image processor, since 1's-counting is rather unnatural for a parallel machine. For the applications of greatest importance, including commercially competitive character recognition, it appeared that a 20 megabit rate was entirely suitable. Since rates higher than this can be achieved even for fairly complex processes with modern components, design of a serial BIP began in earnest. See Fig. 14.

An important function of the BIP is cross correlation; hence it has two image arguments, denoted unknown (U) and mask (M), both taken from core memory. It simultane-

[7] All numbers listed are upper bounds, since simpler measurements might exist for any of the properties. In most cases, it is not likely that such simplifications exist.

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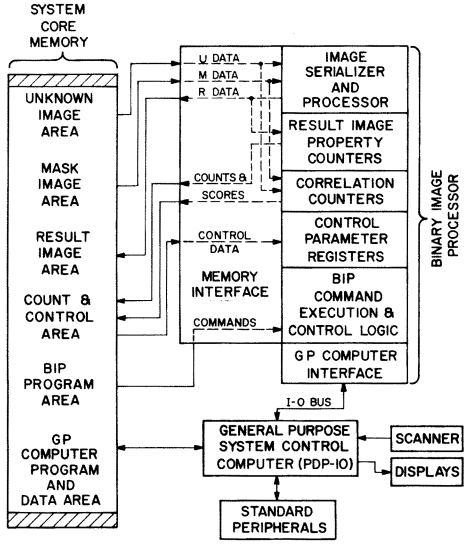


Fig. 14. Binary image processor and its system environment.

ously produces cross correlations at nine relative positions of M with respect to U (corresponding to a 3×3 neighborhood of offsets). Simultaneously, it produces a result image R, which is stored. As it is generated, R feeds a set of local property counters, exactly in the sense of Section VI, which are sufficient for all presently known interesting measurements in two dimensions.

Each bit in the R image is a programmable Boolean function $R_{ij}=R(U_{ij}, M_{ij}, P_{ij})$, where M_{ij} is the ijth bit in the mask image, U_{ij} is the corresponding bit of the unknown U, and P_{ij} is a modified majority-rule function of the neighborhood of U_{ij}, namely $\boldsymbol{U}_{ij}$. To make the P function general, we introduce an eight-bit selection register $S=\{S_0, \cdots, S_7\}$, which is ANDed with $\boldsymbol{U}$. The population of the selected subset of $\boldsymbol{U}$ is denoted $|\boldsymbol{U} \cdot S|$. We now generate the predicate P depending on whether the population number, 0–8, corresponds to a one in a nine-bit "count" register $C=\{C_0, \cdots, C_8\}$. This can be denoted

$$P = C_{|\boldsymbol{U} \cdot S|}. \tag{49}$$

(The subscripts i, j of P and $\boldsymbol{U}$ are omitted for clarity; it is understood that (49) is computed for all i, j.) Thus, the BIP can fatten or thin U in the usual well-known ways [12], using the ordinary neighborhood $\boldsymbol{U}$. As soon as R is completely generated and stored, its local property counts are available for the system's central computer, which is a PDP–10.

The above neighborhood processing structure was the motivation for the neighborhood transformation $\boldsymbol{U} \rightarrow \boldsymbol{U}'$ used in the Euler differential formulas (24), (41)–(43). In the BIPs "Euler mode," $\boldsymbol{U}'$ is used instead of $\boldsymbol{U}$ to compute P, and hence to compute the R image. The selection register can choose the even neighborhood for generating images which are a function of W^*, or the odd neighborhood for Z^*—dependent images, or the entire neighborhood for functions of H^*.

Using the $\boldsymbol{U}'$ operation, the BIP can skeletonize; it can further generate a picture consisting of all the nodes, or all the line ends, or both, of a skeleton. Counting and locating these cells is a powerful form of feature extraction.

The BIP is an autonomous processor, in that it executes its own program once started by the system's general-purpose computer. It can thus sequence itself through a long series of image processes, if desired. For example, it could correlate one unknown against all masks in a typeface stored in core, placing all scores back into memory for further processing by the GP computer, or it could smooth an unknown image, using standard neighborhood majority operations. If the local properties of R, such as area, average line width, and Euler number, as tested by the GP computer, indicate that R is of suitable quality for correlation, the BIP would then change its control bits to start multiple correlation passes. If R still seems defective, more enhancement passes are performed.

Of course, the BIP was not built simply because E, P, and A are locally countable properties. Its design arose out of lengthy experience with other uses of serial image and neighborhood processing. A small BIP was built and used successfully for character recognition in 1968, before we had investigated binary image topology and local property counting. The present, large BIP (about 1600 integrated circuits) has been operational since December 1970.

Acknowledgment

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